

FM-to-AM conversion in high-power lasers

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FM-to-AM conversion is an important issue that could prevent fusion ignition with high-power lasers, such as the Laser MegaJoule (LMJ). We first overview the whole problem of FM-to-AM conversion in high-power lasers and we explain why AM spectral content of FM-to-AM conversion is important, although this information was not used in previous studies. We then propose simple analytical models to simulate FM-to-AM conversion in the LMJ frequency conversion system. We succeed in isolating every cause of spectrum distortion and give, for each of them, FM-to-AM predictions that are in very good agreement with simulations of a complex propagation code. Finally, we show how the last grating filters most of the FM-to-AM conversion. We conclude that the FM-to-AM conversion distortion criterion will be, on LMJ, below 40% in the last optics and 10% on the target. © 2008 Optical Society of America
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1. Introduction

The Laser MegaJoule (LMJ) in France and the National Ignition Facility (NIF) in the United States are high-power laser facilities designed to achieve inertial confinement fusion (ICF) of a deuterium-tritium (D-T) capsule (the target) [1]. The control of the temporal shape of the laser pulse is a key point to ensure an optimal interaction of laser-target. The main difficulty in controlling the temporal shape is the FM-to-AM conversion. Indeed, for propagation purposes, the optical spectrum has to be broadened thanks to phase modulation, i.e., frequency modulation (FM). Ideally, this pure phase modulation does not change beam intensity. However, propagation through the different components of the chains is not ideal and slightly filters the optical spectrum. Therefore, FM is partly converted into intensity modulation, i.e., amplitude modulation (AM) (Fig. 1) [2]. This intensity modulation needs to be reduced or at least controlled for different reasons. First, the target

needs perfect symmetry of irradiation; the same intensity at each moment for each of the 240 LMJ beams is required. For this reason, AM can prevent the ignition of the target. Hydrodynamic response of the target for this symmetry is about 10 GHz [3], so the low AM frequencies must be reduced first. Furthermore, AM can amplify instabilities of the plasma (back Raman and Brillouin scattering) around the target; since the plasma response is very fast ($\sim$ ps), there is no frequency limit for the control of the intensity modulations [3]. Finally, intensity peaks due to AM might also damage the optics and, so, the laser performance.

A LMJ beam is composed of different specific stages, as presented in Fig. 2: the source that shapes the pulse temporally and spectrally; the preamplifier module (MPA), which is the first stage of amplification (gain of 10^9) that also shapes the pulse spatially; the amplifier chain that gives the required energy; and, finally, the frequency conversion and focusing system that converts IR at a 1053 nm wavelength to UV at a 351 nm wavelength for a better interaction with the hohlraum [3]. The optical spectrum of the

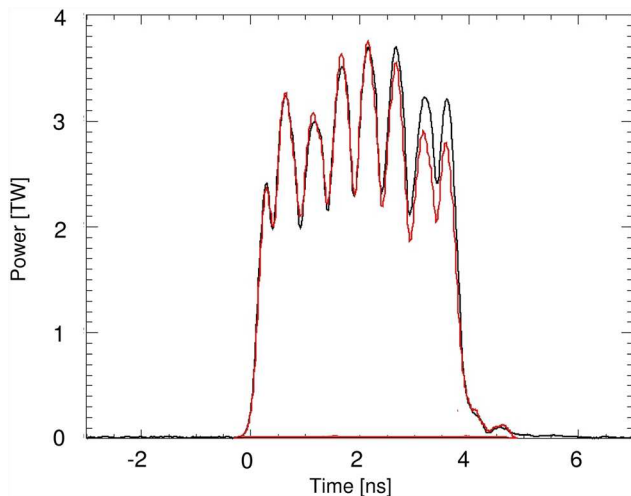


Fig. 1. (Color online) Temporal measurement of the power on the LIL, prototype of the LMJ (black curve and red curve are from two different detectors). The beam was initially spectrally shaped with a phase modulation at 2 GHz, $m = 7$ rad. Because of FM-to-AM conversion, the initial flat-top pulse shape is modulated.

laser pulse may be filtered at almost every stage of the chain; each component with a nonuniform spectral transmission may convert FM to AM. FM-to-AM conversion first appears during the propagation over polarization-maintaining fibers (PMF) in the source [2]. The resulting AM used to be important, but this problem can now be managed [4]. In the MPA, part of FM is converted into AM because of the effects of coatings, gain narrowing, and interferences. We can compensate for the distortion of the spectrum by use of a tunable filter, such as a Lyot filter, that can be implemented before the preamplification stage [5]. Indeed, all the effects that distort the spectrum are linear phenomena and can be considered as linear transfer functions and precompensation induces minimum loss of energy. For opposite reasons, in the amplification chain, intensity modulation cannot be easily pre- or postcompensated. However, there is almost no FM-to-AM conversion since the gain is sufficiently wide in the spectrum domain and does not filter the signal. Hence, most of the expected FM-to-AM conversion is due to the final optics of the chain, including gratings and crystals for frequency conversion. As a matter of fact, frequency

conversion is performed with KDP crystals, the only suitable crystals for such lasers; KDP crystals may have a large section ($40 \text{ cm}^2 \times 40 \text{ cm}^2$), but they have a relatively narrow spectral acceptance [6]. Moreover, gratings are known to induce a spectral distortion, such as in chirped-pulse amplification systems [7].

In the following sections, we present in detail FM-to-AM conversion and focus on the importance of AM spectral content. Some experimental results on LIL, the LMJ prototype in Section 2, are also given. Then, we study FM-to-AM conversion in the frequency conversion systems through both analytical models and numerical simulations. Finally, we propose a model for AM filtering in the focusing system (Section 3). Throughout this paper, numerical simulations are realized with the CEA's homemade Miró propagation code [8].

2. Interests and Issues of Phase Modulation

A. Phase Modulation Overview

LMJ pulses are phase modulated for two reasons. First, levels of spectral power density (SPD) for a not spectrally broadened ignition laser pulse would exceed transverse the Brillouin threshold in optics at the end of the amplification chain. Such stimulated Brillouin scattering (SBS) would strongly damage these optics [9]. Anti-Brillouin phase modulation spreads energy to different wavelengths (frequencies) in order to lower the SPD maximum level below the SBS threshold. Second, focal spots on the LMJ require a high spatial homogeneity. This is realized by generating a time-varying speckle pattern by using a spectrally broadened pulse and a dispersion system (grating) that separate the different speckle patterns of different wavelengths. This scheme is called smoothing [10]. For the LMJ, we use two different sinusoidal phase modulations to solve both difficulties. A sinusoidal phase modulation is defined by two parameters: the modulation frequency f_m and the modulation index m . To determine the associated optical spectrum, we consider that the initial low frequency temporal shape of the pulse is negligible. Therefore, for a unique sinusoidal modulation the optical field is given by

$$a_0(t) = \exp[i m \sin(2\pi f_m t)]. \quad (1)$$

This is a periodic signal and the corresponding optical spectrum is composed of uniformly separated f_m spaced Dirac peaks:

$$\tilde{a}_0(f) = \sum_{n=-\infty}^{+\infty} J_n(m) \delta(f - n f_m), \quad (2)$$

where $J_k(m)$ is the k th Bessel function [11]. The spectrum has theoretically an infinite width, but 98% of the signal energy is included between $\Delta f = 2(m+1)f_m$ (Carson's rule) [12].

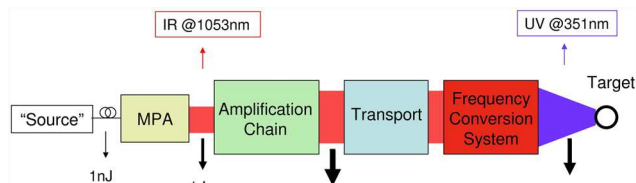


Fig. 2. (Color online) Functional diagram of an LMJ beam. The laser pulse is shaped temporally and spectrally in the "source." Then the pulse is amplified in the MPA and in the amplification chain. The transport section is composed of dielectric mirrors. Its function is to propagate the beam to the SCF that converts the beam into UV light. The SCF also focuses the beam to the target.

We use a modulation at 2 GHz with a modulation index of 7 to avoid SBS that creates a 32 GHz bandwidth spectrum that is not enough to smooth the beam [Fig. 3(a)]. Thus, when smoothing is required, we add a second phase modulation at 14.25 GHz with a modulation index of 5 [Fig. 3(b)]. The choice of smoothing modulation is given by the grating features [10]. When both phase modulations are applied, the spectrum is the convolution of both of them and the spectral width is 201 GHz (32 GHz + 171 GHz), i.e., 0.75 nm at the central wavelength of 1053 nm. After frequency conversion, the width is multiplied by 3, ~600 GHz.

B. FM-to-AM Conversion

Phase modulation does not change the intensity shape I because $I(t) = |a_0(t)|^2$, but if any optical device during propagation changes the optical spectrum (so the optical field by inverse Fourier transform), it will modulate the intensity; this is FM-to-AM conversion. Note that FM-to-AM conversion has been initially used for FM demodulation in telecommunications [13]. To quantify the FM-to-AM conversion, the distortion criterion α defined by the following expression is used:

$$\alpha = 2 \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}}, \quad (3)$$

α ranges from 0 to 200% and is ideally equal to 0 (no intensity variation).

Most of the contributions of FM-to-AM conversion are linear (some other cases are studied in the second part of this paper in detail) so we use a transfer function approach by working in the Fourier transform space. But because of the nonlinear link between optical field and intensity, analytical calculations of the resulting AM are difficult as well as detection of extrema. For a simple distortion, such as group velocity dispersion (GVD) or gain narrowing, good approximations can be done. These results and complements

are summed up in Table 1. We refer to [4] for more details in calculations.

C. Spectral Content of AM for a Single Sinusoidal Phase Modulation

For the same α value, different AM spectral contents can have different effects on laser damage and on laser–target interaction. Hence, the only scalar value α is not sufficient. Note that AM due to optical spectrum distortion is also formed of f_m -spaced peaks from the lowest frequency f_m to the highest possible Δf . But the spectral content depends on the phase modulation and on the transfer function. Figure 4 shows two AM spectra for different transfer functions with about the same α value. In the following, we compute the AM spectrum for a single sinusoidal phase modulation.

Let us consider a filtering function (that can be complex) $H(f)$. After the filter, the intensity spectrum $I_{\text{out}}(f)$ is given by (see Appendix for calculations)

$$\tilde{I}_{\text{out}}(kf_m) = \sum_{n=-\infty}^{+\infty} H^*(-nf_m)H \times [(k-n)f_m] J_{-n}(m) J_{k-n}(m). \quad (4)$$

For any transfer function, Eq. (4) does not bring a simple result and numerical applications are necessary. Usually, the first harmonics of AM are the highest and have the most detrimental effect on the laser–target interaction [3]. Hence, shifting the AM spectrum toward higher harmonics may be of interest. Therefore, we concentrate on the first harmonic, but considering all odd harmonics is no more difficult. If k is an odd number,

$$\tilde{I}_{\text{out}}(kf_m) = (-1) \sum_{n=-\frac{(k-1)}{2}}^{+\infty} \{H^*(nf_m)H[(n+k)f_m] - H^*(-nf_m)H[-(n+k)f_m]\} J_n(m) J_{n+k}(m). \quad (5)$$

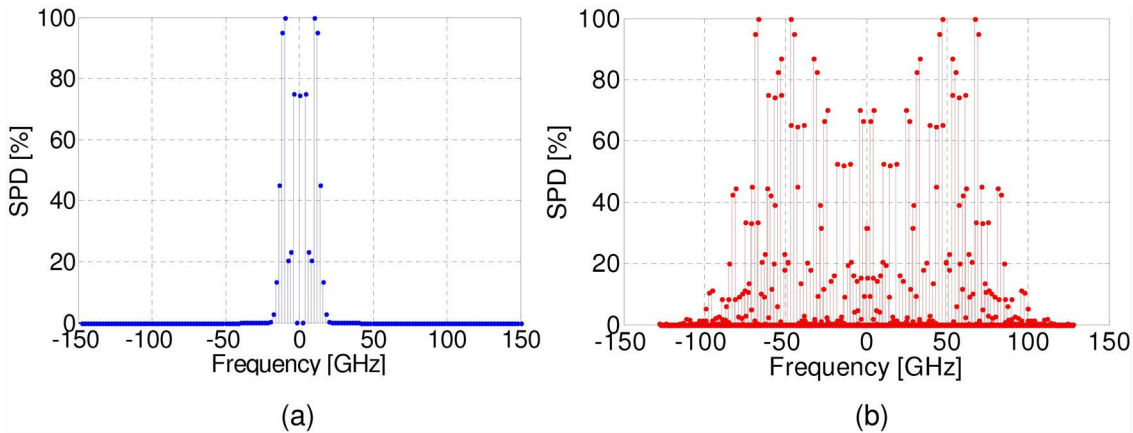


Fig. 3. (Color online) (a) Optical spectrum for the 2 GHz sinusoidal phase modulation with an index m of 7 (also called anti-Brillouin modulation). (b) Optical spectrum for both modulations used on the LMJ: anti-Brillouin modulation and smoothing modulation at the frequency of 14.25 GHz and a modulation index of 5.

Table 1. Examples of Analytical Values of Distortion Criterion α Due to Simple Filters for a Sinusoidal Phase Modulation at Frequency f_m with Modulation Index m^a

Transfer Function	α value	
	$ f_c < mf_m$	$ f_c > mf_m$
$1 + i\sigma(f - f_c)$	$\alpha = \frac{\sigma^2 f_c + mf_m ^2}{1 - 1/2\sigma^2 f_c + mf_m ^2}$	$\alpha = \frac{4\sigma^2 m f_m f_c}{1 - \sigma^2 f_c^2 + (mf_m)^2 ^2}$
$1 - \sigma(f - f_c)^2$	$\alpha = \frac{2\sigma f_c + mf_m ^2}{1 - \sigma f_c + mf_m ^2}$	$\alpha = \frac{8\sigma m f_m f_c}{1 - 2\sigma f_c^2 + (mf_m)^2 ^2}$
$1 - i\sigma(f - f_c)^2$	$\alpha = 4\sigma m f_m^2$	

^aThese transfer functions can be used to simulate GVD, gain narrowing, or spectral dispersion of the gratings.

These components are null if

$$\forall n \geq -\frac{(k+1)}{2}, \quad H^*(nf_m)H[(n+k)f_m] - H^*(-nf_m)H[-(n+k)f_m] = 0. \quad (6)$$

The Hermitian or anti-Hermitian functions H defined with the following property verify Eq. (6):

$$\forall x \in \mathbb{R}, H^*(-x) = \pm H(x). \quad (7)$$

So for all these transfer functions, there are no odd harmonics (not even the fundamental) of f_m in the AM spectrum. It means that the first frequency of the AM spectrum will be $2f_m$ and not f_m . This kind of transfer function is usual; gain narrowing around the central wavelength satisfies Eq. (7) and, therefore, there are only even harmonics of AM.

We exhibit this result in Fig. 5 thanks to a parametrical interferometric filter. The latter is generated by a fibered two-wave interferometer composed of a first polarizer at 45° of the fiber axis, a 10 m long polarization-maintaining fiber, and a second polarizer. Its transfer function is expressed by

$$H(f) = 1 + A \exp[i(2\pi \cdot f \cdot \Delta\tau + \psi)], \quad (8)$$

where $\Delta\tau = 15$ ps is proportional to the fiber length, $C = 2A/(1 + A^2) = 0.9$ is the contrast that is determined by the angle of the second polarizer, and ψ is the recombination phase determined by constraints applied on the fiber [4]. With $\psi = 0^\circ$, H satisfies Eq. (6) [Fig. 5(a)]. Thus, only even harmonics should appear. This is confirmed experimentally: we applied on the signal the anti-Brillouin phase modulation at 2 GHz and the FM-to-AM conversion has been measured for $\psi = 0^\circ$ in the time domain with a 6 GHz oscilloscope. The AM spectrum has been computed with the recording traces [Fig. 5(b)] and only 4 GHz intensity modulation ($2f_m$) exists in the AM spectrum (6 GHz is an artifact since we used a 6 GHz bandwidth oscilloscope).

D. Spectral Content of AM on the Laser MegaJoule

Up to now, we considered a single sinusoidal phase modulation. However, we use two different sinusoidal phase modulations at 2 GHz and 14.25 GHz. Thus, an LMJ pulse will have a large AM spectral content, larger than what can be measured because of the low cutoff frequencies of detection systems. The following example, met on the Laser Integration

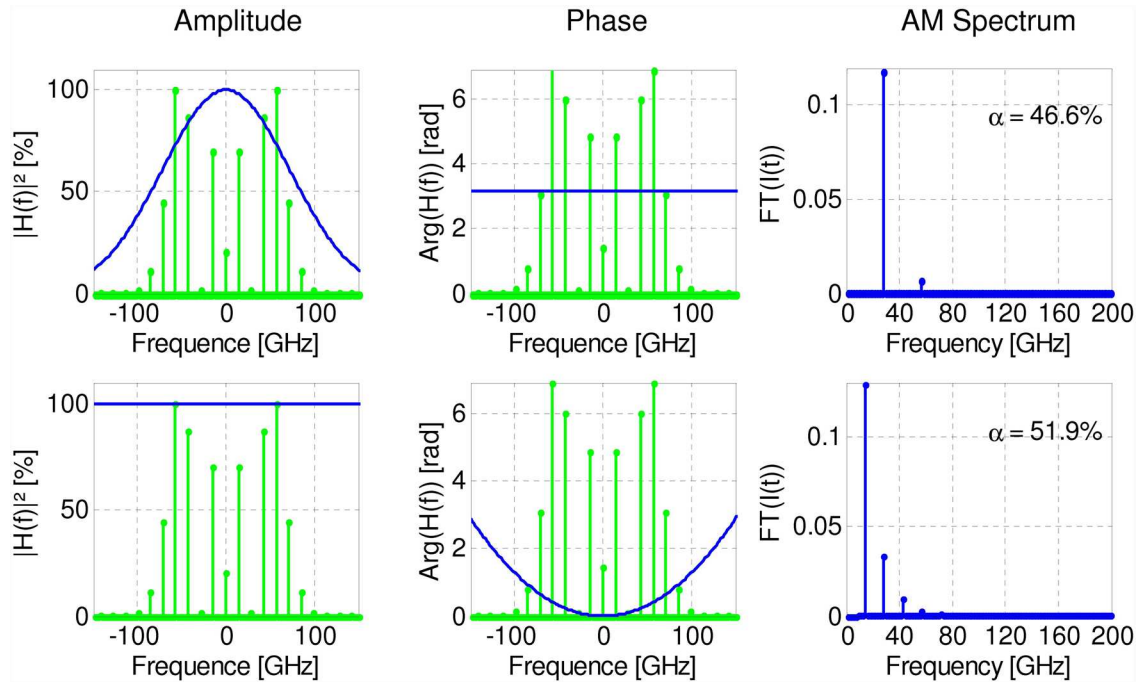


Fig. 4. (Color online) Theoretical transfer function H (in blue) applied on a phase modulation (in green with $f_m = 14.25$ GHz, $m = 5$). Top line: Gaussian bandpass filter with a 170 GHz FWHM (see Section 3.B for another similar filter). Bottom line: Chromatic dispersion. For example, dispersion in a 200 m long fiber at 1053 nm (dispersion parameter of -11 ps/nm) (see Section 3.C for another similar filter). The two AM spectra are totally different, although the resulting values of distortion criterion α are about the same.

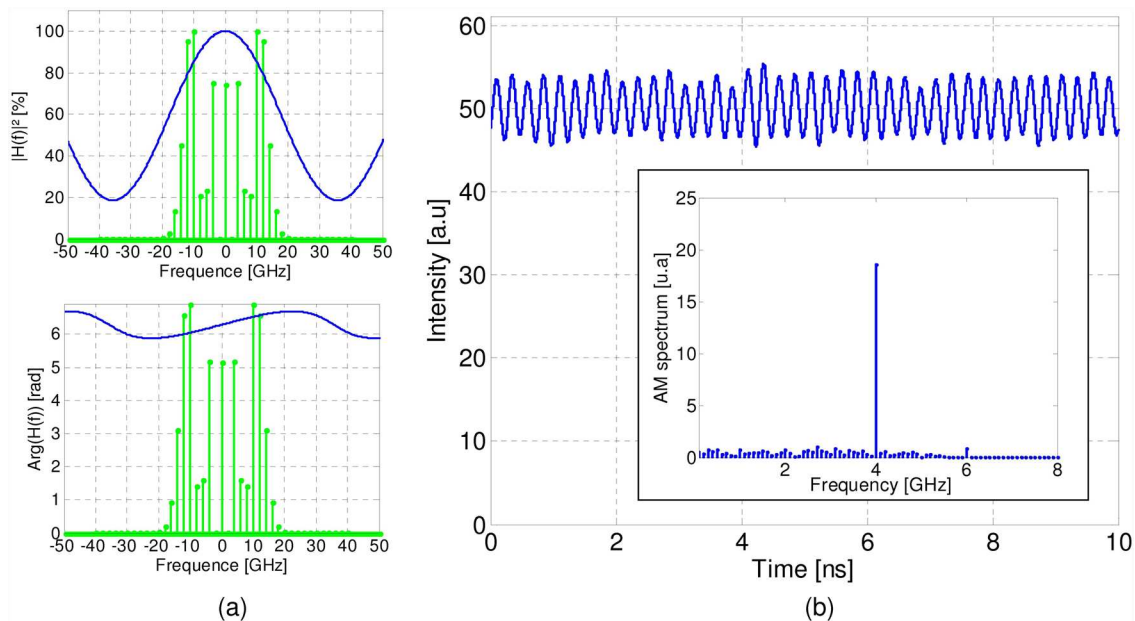


Fig. 5. (Color online) (a) Anti-Brillouin modulation ($f_m = 2$ GHz, $m = 7$) is filtered by a Hermitian transfer function. (b) Resulting AM measured (insets: AM spectrum associated) with a 5 GHz photodiode and a 6 GHz bandwidth oscilloscope. There is no AM at 2 GHz; because of the filter, only even harmonics of frequency modulation can exist (here, 4 GHz).

Line (LIL) facility, the prototype of LMJ, is explicit about this difficulty. Anti-Brillouin modulation is always activated, while smoothing modulation may be activated or not. Values of the distortion criterion α on simple transfer functions show that, in a real case, it is expected that the broader the pulse spectrum, the higher the level of the α factor. This is also what is observed with numerical simulations of the laser chain. We measured the intensity modulation (AM) of 23 consecutive shots on LIL. As can be seen in Fig. 6, the FM-to-AM conversion level α is fluctuates highly from one shot to another. This feature is well known and is due to the propagation of the pulses in the polarization-maintaining fibers in the source of the laser and has been solved after the measurements [5]. The full square shots in Fig. 6 correspond to anti-Brillouin only, while the empty square shots correspond to both anti-Brillouin and smoothing. The measured values of α are among the lowest when smoothing is activated. Actually, single shot measurements are limited to a few gigahertz (typically 6 GHz). This explains why the values of α with smoothing (full square shots) are not necessarily among the highest. To explain why they are among the lowest, we have to consider Fig. 7: anti-Brillouin and smoothing modulations are applied and the optical spectrum is the convolution of the optical spectrum of the anti-Brillouin spectrum alone and the smoothing spectrum alone. So, considering one peak of the anti-Brillouin spectrum, the smoothing creates replicas that are filtered by a different part of the transfer function. Therefore, by averaging between replicas, FM-to-AM conversion is reduced for the AM frequency due to anti-Brillouin modulation [14].

An important conclusion is that it is possible to create preferentially AM at high frequencies; with two

modulations, most of the AM will be at the higher modulation frequency (for the LMJ, instead of a 2 GHz AM, we will have first 14.25 ± 2 GHz AM). For laser–target interaction, the most detrimental frequencies are the lowest. The whole AM spectrum must be considered for the plasma around the target. However, we will present in Section 3.E that the frequencies beyond 7 GHz are considerably filtered. So, possibilities to create preferentially AM at high frequencies is interesting to reduce the impact of FM-to-AM conversion. In the following section, we estimate the FM-to-AM conversion in high-power lasers such as the LMJ and we point out the impact of the AM spectral content in that case.

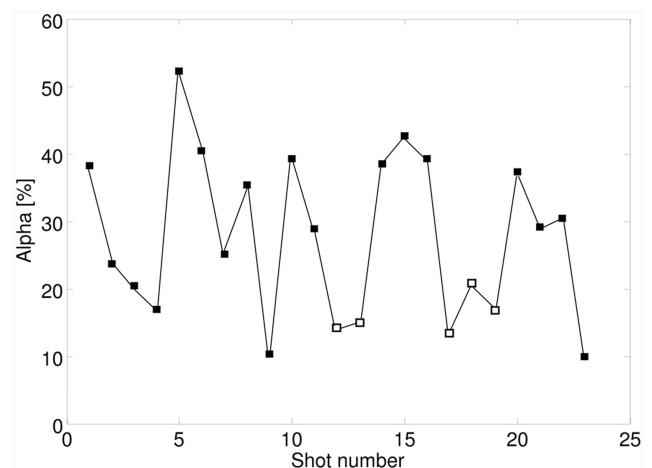


Fig. 6. Experimental results of FM-to-AM conversion on 23 consecutive shots on LIL. The measured values of α are among the lowest when anti-Brillouin (solid squares) and smoothing (empty squares) phase modulations are both activated, although FM-to-AM is supposed to be higher for a larger spectrum.

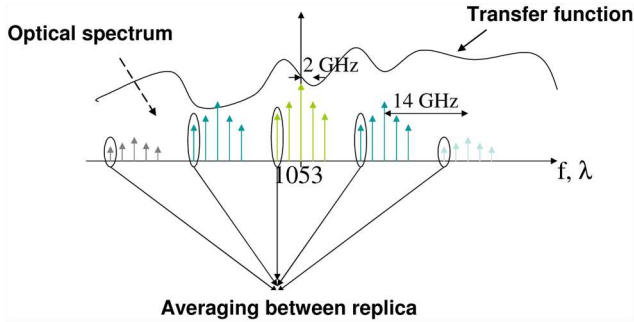


Fig. 7. (Color online) AM is reduced at 2 GHz when the spectrum is broadened with the second phase modulation at 14.25 GHz.

3. Filtering Functions and FM-to-AM Conversion on the Laser MegaJoule

Since FM-to-AM conversion of the source has been managed, the conversion of the MPA can be compensated for, and the conversion of the amplification chain and transport section is negligible, we focus our attention on the frequency conversion and focusing system characteristics of high-power lasers and we give results on the LMJ configuration.

A. Laser MegaJoule Frequency Conversion System

After the amplifier section, the IR beams at 1053 nm (1ω) are transported to the frequency conversion and focusing system (SCF) to be converted into UV at 351 nm (3ω) thanks to two nonlinear crystals of KDP and DKDP as presented in Fig. 8. Each UV beam has to deliver about 7.5 kJ so the energy conversion efficiency has to be as good as possible (i.e., $\sim 80\%$ at maximum intensity). Phase modulated spectra for smoothing are larger than the spectral acceptance of the frequency conversion system. To improve the conversion efficiency, a plane grating has been added before the crystals; the grating angularly disperses the optical frequencies of the FM spectrum to compensate for most of the phase mismatch [6,15]. The plane grating also creates an optical chicane just before the target. Hence, the optics avoid a direct exposure to the neutrons created (see Fig. 8). Finally, to

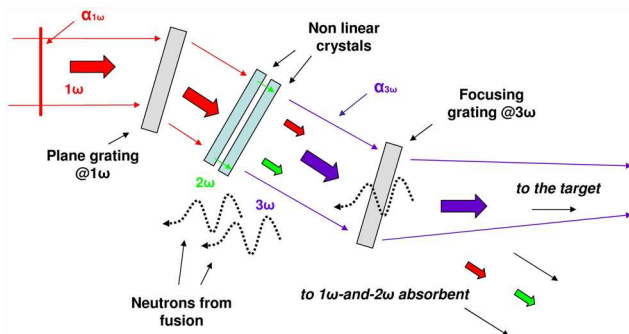


Fig. 8. (Color online) Simple description of the SCF on LMJ. Crystals convert IR into UV. The plane grating improves efficiency conversion by dispersion; phase mismatch over the spectrum is considerably lowered. The focusing grating separates UV light from the first and second harmonics and it focuses the beam to the target.

focus the beams and to get rid of the unwanted 1 and 2ω residue, a focusing grating is used.

In the following, we separately consider the impact on FM-to-AM conversion of the nonlinear crystals (Section 3.B), and the one of the plane grating at 1ω (Section 3.C). In Section 3.D, we combine the effects of both components. And finally, we consider the influence of the focusing grating at 3ω (Section 3.E).

B. AM Induced by Frequency Conversion

FM-to-AM conversion due to frequency conversion has two different origins: the amplification of AM already existing at 1ω and the phase mismatching in the nonlinear crystals that induces a transfer function. Amplification of the intensity modulation from the previous stages is a direct consequence of the nonlinearity of the frequency conversion. Let us consider the following example of a low intensity third harmonic generation (THG) in the case of a perfect phase matching over the entire spectrum. The intensity after frequency conversion is given by $I_{3\omega} \propto I_{1\omega}^\beta$, where β is a constant equal to 3 for low intensity cases [8]. So, for small intensity variations at 1ω due to AM, $I_{1\omega} = I_{1\omega}^0(1 + \epsilon)$ so $I_{3\omega} \propto (I_{1\omega}^0)^\beta(1 + \epsilon)^\beta \approx I_{3\omega}^0(1 + \beta\epsilon)$ that we can identify with AM at 3ω . Thus,

$$\alpha_{3\omega} = \beta\alpha_{1\omega}. \quad (9)$$

AM already existing before the frequency conversion is amplified by a coefficient linked to the nonlinearity of the frequency conversion. For higher intensity levels $\beta < 3$ because of frequency conversion saturation; β is intensity dependent and with Miró we have estimated that β ranges between 1.30 at 3 GW/cm^2 and 1.49 at 1 GW/cm^2 on LMJ. As long as $\alpha_{1\omega}$ is sufficiently small, β can be considered as constant and Eq. (9) is accurate enough, as can be seen in Fig. 9.

The second origin of FM-to-AM conversion is the optical spectrum distortion due to phase mismatching. The spectral acceptance of the nonlinear crystals is limited due to group velocity differences inside crystals, which generates AM. Phase mismatching prevents a good conversion efficiency conversion, which led to a choice of a limited optical spectrum width; frequencies in the optical spectrum are still not perfectly converted, except the central wavelength, and it affects FM-to-AM conversion. Indeed, it can be interpreted as a linear filtering of the theoretical 3ω spectrum that could have been converted if there was perfect phase matching over the entire spectrum. Indeed, let us consider the classical equation of frequency conversion for THG in the low-intensity case without plane grating in front [8]:

$$A_3(\Delta k) = \frac{A_{3\max}}{L} \frac{e^{i\Delta k L} - 1}{i\Delta k} = A_{3\max} e^{\frac{i\Delta k L}{2}} \text{Sinc}\left(\frac{\Delta k L}{2}\right), \quad (10)$$

where A_3 is the optical field at 3ω , Δk is the phase mismatching, and L is the crystal thickness.

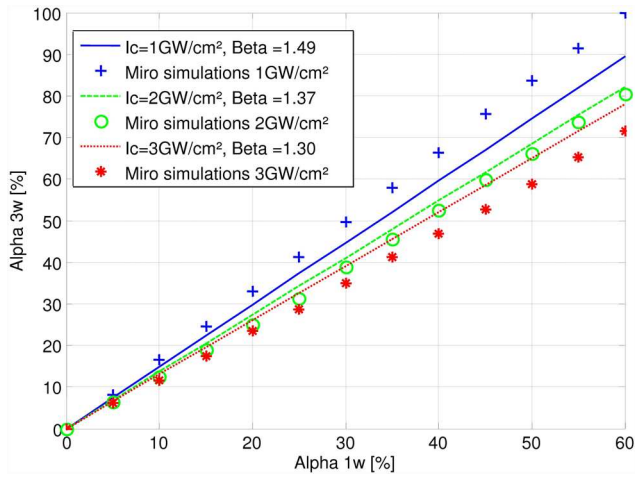


Fig. 9. (Color online) Miró simulations for FM-to-AM conversion in THG (case of a perfect phase matching overall the spectrum). Amplification of AM is always verified, but the value of amplification decreases with the intensity. Curves are predictions with the model $a_{3\omega} = \beta a_{1\omega}$ where β is defined by the fact that $I_{3\omega} \propto I_{1\omega}^\beta$. Differences come from the fact that β instantaneously depends on the intensity.

Assuming that the phase mismatching is proportional to the frequency, frequency conversion with phase mismatching can be interpreted as a linear filter of A_3 . With the same hypothesis, the complex exponential in Eq. (10) is a time delay and does not create AM. So AM resulting from phase mismatching is only due to the second part, which is the spectral acceptance of frequency conversion for the low-intensity case and we write the effective transfer function for FM-to-AM conversion as

$$H_c(f) = \text{Sinc}[\gamma(f - f_c)] \approx 1 - \frac{\gamma^2}{6}(f - f_c)^2, \quad (11)$$

with f_c as the phase matching frequency and

$$\gamma = \frac{L}{2} \left(\frac{\partial \Delta k}{\partial f} \right)_{3\omega}.$$

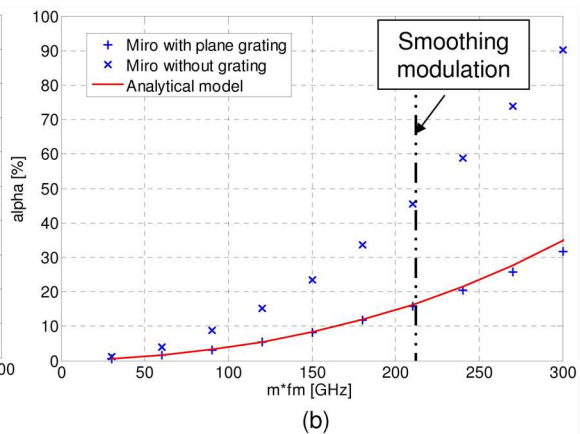
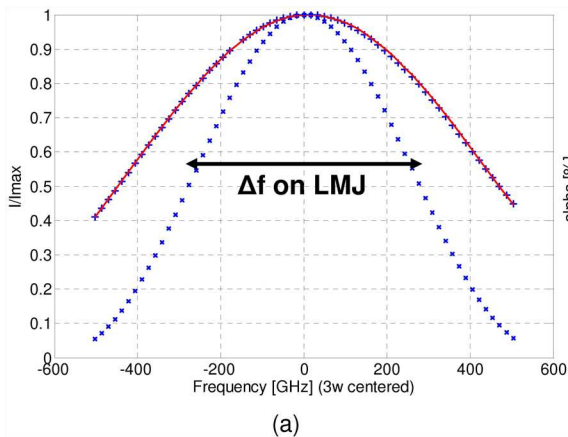


Fig. 10. (Color online) (a) Spectral acceptance for the frequency conversion doubler-tripler system at 2 GW/cm^2 . x: case without plane grating and +, case with plane grating (LMJ). (b) AM resulting from spectral acceptance transfer function. Red curve corresponds to analytical formula results according to Eq. (12).

According to Table 1 and noting that the phase modulation index of the converted wave A_3 is 3m, the distortion criterion α_c induced by H_c is

$$\alpha_c = \frac{\frac{\gamma^2}{3}(f_c + 3mf_m)^2}{1 - \frac{\gamma^2}{6}(f_c + 3mf_m)^2}. \quad (12)$$

In the following, we show that this result is still valid for less restrictive assumptions than THG in the low-intensity case and can be used for LMJ configuration. Indeed, high intensities and the plane grating dispersion change mainly the spectral acceptance given by γ [6]. Because of the complex LMJ configuration, γ cannot easily be determined and we choose numerical simulations to estimate it. Figure 10(a) shows the spectral acceptance for LMJ at 2 GW/cm^2 with plane grating. We have determined the associated parameters γ and f_c from Eq. (11) by fitting the acceptance spectral curve given by Miró. To verify if the analytical formula in Eq. (12) is a good approximation, we have plotted in Fig. 10(b) α_c versus mf_m ; numerical simulations of the whole system with Miró are in very good agreement with our analytical model.

When smoothing modulation is applied, contribution from spectral acceptance leads to $\alpha_c = 17\%$. We have also considered another configuration without plane grating. In this configuration, the spectral acceptance is much reduced and α_c increases to 47%. Also, the frequency conversion efficiency is lowered for the same reason. Therefore, the improvement brought by the plane grating seems important but we have also to consider in the following section the impact on FM-to-AM conversion of the grating by itself.

C. AM Induced by Plane Grating

The plane grating converts FM to AM by propagation due to spatial dispersion. This grating is used to spatially separate the different wavelengths. So the accumulated phase after the grating is different from one wavelength to another. The frequency

dependence of the phase after a plane grating due to dispersion is well known since it is used in compressors for ultrashort pulses [16] and the accumulated phase φ can be written as follows:

$$\varphi(f) = \varphi_0 + \varphi_1 f + \frac{1}{2} \varphi_2 f^2 + o[f^2]. \quad (13)$$

Thus, the optical field is linearly filtered by $\exp[i\varphi(f)]$ and converts FM to AM. But the first two terms, φ_0 and $\varphi_1 f$, do not bring temporal distortions because φ_0 is a constant and $\varphi_1 f$ is a temporal delay. So the useful part of the transfer function is given by

$$H_d(f) \approx \exp\left[\frac{i}{2} \varphi_2 f^2\right] \approx 1 + \frac{i}{2} \varphi_2 f^2. \quad (14)$$

For a propagation distance x after a plane grating in the case of a Littrow configuration at the angle θ_0 , we have for the central frequency f_0 , with a density of grooves N [17]:

$$\varphi_2 = -\frac{2\pi c N^2 x}{f_0^3 \cos^2(\theta_0)}. \quad (15)$$

According to Table 1, the distortion criterion α_d induced by H_d is

$$\alpha_d(D) = 2|\varphi_2(x)|m f_m^2 = \frac{4\pi c N^2}{f_0^3 \cos^2(\theta_0)} m f_m^2 x. \quad (16)$$

On LMJ, $\theta_0 = 25^\circ$, $\lambda_0 = 1053 \text{ nm}$, $N = 802.7 \text{ mm}^{-1}$, and, for smoothing modulation, we find that the distortion criterion is proportional to the propagation distance x with a coefficient of $13\% \text{ m}^{-1}$. This analytical result is confirmed by Miró simulations as shown in Fig. 11.

D. Combined Effects of Plane Grating and Frequency Conversion

Since frequency conversion and propagation after the plane grating convert FM to AM, we now consider the combined FM-to-AM conversion. We have shown in Sections 3.B and 3.C that two independent phenomena distort the FM spectrum, leading to two different AM spectra composed of multiple frequencies with no phase relation. As a matter of fact, Eq. (11) satisfies Eq. (6), thus, the AM spectrum due to frequency conversion is composed only of even harmonics. On the contrary, Eq. (14) does not satisfy Eq. (6) and the AM spectrum due to plane grating starts at f_m . Therefore, we assume that AM created by the plane grating and AM created by the frequency conversion can be considered as two independent noises. Thus, the resulting distortion criterion α_{tot} can be estimated by the root mean square of each contribution and the distortion criterion α_{sum} for a distance x after the plane grating is

$$\alpha_{\text{sum}}(x) = \sqrt{\alpha_c^2 + \alpha_d^2(x)}. \quad (17)$$

In practice, the nonlinear crystals are separated from the plane grating by a distance l (about 50 cm dependent of the beam considered on LMJ). Thus, to give a more accurate model of the whole FM-to-AM conversion in the frequency conversion system before focusing the beam, we modify Eq. (17), noticing that the AM generated by propagation after the plane grating is amplified in the nonlinear crystals according to Eq. (8), and it leads to the distortion criterion α_{NF} for the near field at a distance x after the nonlinear crystals:

$$\alpha_{\text{NF}}(x) = \sqrt{[\beta \alpha_d(l)]^2 + \alpha_c^2 + \alpha_d^2(x-l)}. \quad (18)$$

We have plotted Eq. (18) in Fig. 11. Numerical simulations with Miró are once again in close agreement with our analytical model, confirming the assumption that the different FM-to-AM contributions can be considered as independent noises. On LMJ, the propagation distance between the plane grating and the focusing grating is about $D = 2 \text{ m}$. Hence, the combined effect of the plane grating and the frequency conversion yields $\alpha_{\text{NF}} = 28\%$. This value is comparable to the 47% that is obtained for frequency conversion without grating (see Section 3.B). It is even possible to decrease the FM-to-AM conversion by setting the focusing grating closer to the crystals to reduce D . However, KDP crystals would receive a direct exposure of neutrons that would be created in the fusion of the target.

We have established different models that explain the origins of FM-to-AM conversion in the SCF and give good results in the case of single phase modulation. To complete this study, we must consider that,

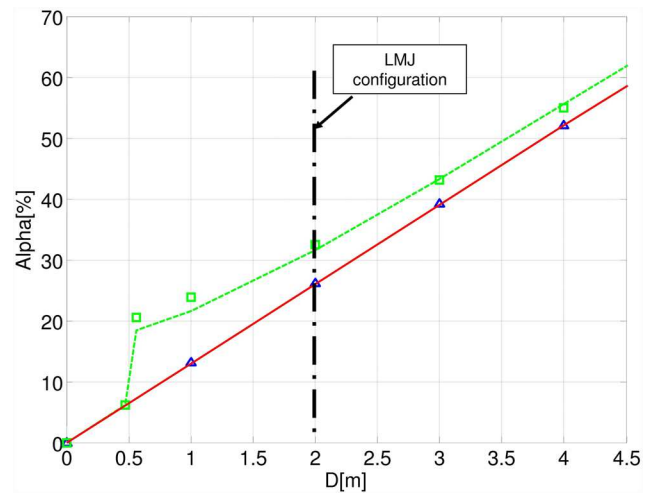


Fig. 11. (Color online) FM-to-AM conversion due to propagation after 1ω plane grating: Miró simulations (blue triangles) versus analytical model according to Eq. (16) (red curve) for a phase modulation $m = 5 \text{ rad}$ at 14.25 GHz . Distortion criterion is proportional to the distance propagation with a coefficient of $13\% \text{ m}^{-1}$. Green squares consider also the effects of crystals on FM-to-AM conversion. Green dashed curve is the analytical model according to Eq. (18) considering the addition of different effects: for long propagation distance, the crystals' effect is negligible.

on LMJ, we are using two different modulations: smoothing modulation (used for simulations) and anti-Brillouin modulation at 2 GHz with $m = 7$. Spectrum enlargement brought by this second modulation is about 20% and the anti-Brillouin frequency modulation is considerably smaller than the smoothing frequency modulation. With both of these considerations, we estimate the influence of anti-Brillouin modulation on FM-to-AM conversion. First, the impact on propagation between the gratings is almost negligible. Indeed, we have shown that α_d is proportional to mf_m^2 , so a 2 GHz modulation is almost negligible with a 14.25 GHz modulation. Results of Miró simulations summed up in Table 2 confirm this assumption: α_d is the same with or without anti-Brillouin modulation. The worst effect is the spectral acceptance transfer function because α_c increases with $m^2 f_m^2$ [see Eq. (12)], i.e., with Δf^2 . In Table 2 we can see that α_d increases from 17% to 24%, justifying our explanation since $24 \approx (1 + 20\%)^2 \times 17$. Finally, for both modulations, with plane grating and propagation, we have $\alpha_{NF} = 35\%$ (28% with only smoothing modulation).

Such levels of FM-to-AM conversion are not acceptable for a good ignition of the target. But the target is in the focal plane of the focusing grating. Therefore, we have to consider the contribution of the focusing grating on FM-to-AM conversion.

E. AM Reduction Induced by Focusing Grating

The focal spot of high-power lasers for fusion needs a high homogeneity over its millimetric width. For this purpose, a phase plate shapes the focal spot and the phase modulation is coupled with a dispersion system (focusing grating on LMJ) to smooth the beam, i.e., to spatially average the intensity [3]. On LMJ, longitudinal smoothing by spectral dispersion (LSSD) is used; another classical scheme is transverse smoothing by spectral dispersion (TSSD). Both configurations use gratings: plane or focusing, as presented in Fig. 12. Gratings are interferential systems that privilege one direction for the equiphase surface, which is different from the isointensity surface after the grating, as shown in Fig. 12. Therefore, in both TSSD and LSSD configurations, energy in the laser pulse has a time delay dependent on the spatial location.

Table 2. Distortion Criterion for Frequency Conversion Systems: Miró Results for Different Configurations^a

	14.25 GHz		14.25 + 2 GHz	
	without grating	with grating	without grating	with grating
α_c	47%	17%	66%	24%
α_d	0%	26%	0%	27%
α_{NF}	47%	34%	66%	39%
α_{FF}	3%	8%	12%	9%

^a α_c is the contribution of nonlinear crystals. α_d is the effect of grating dispersion. We consider also α_{tot} , the distortion criterion at the focusing grating surface, and $\alpha_{tot,filtered}$ is the value seen in the focal plane. Results for the LMJ configuration are in the fourth column.

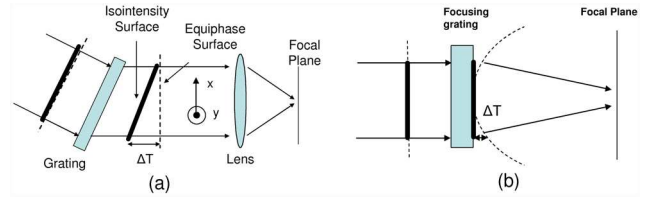


Fig. 12. (Color online) Simple schemes of (a) TSSD and (b) LSSD (LMJ configuration). In both cases, integration of intensity is due to time delay ΔT , so AM is lowered in the focal plane.

Let us consider TSSD first. The plane grating generates a time delay proportional to x (Fig. 12). Then the lens simply acts as an integrator in the focal plane [18]. Indeed, the different parts of the waveform at a given time on the lens surface arrive on the focal plane at a related given time after. (Note that a lens does not introduce significant time delay [17].) It means that the far-field intensity is temporally averaged due to the time delay and if we have a temporal waveform distorted by FM-to-AM conversion, AM is lowered by temporal integration. Intensity summation is justified by the use of the phase plate, making the beam spatially incoherent. Although more complex, the phenomenon is the same on LMJ: the focusing grating both generates the spatially varying time delay and its focusing function leads to the intensity summation for the far field.

Using this idea that has already been noted in the TSSD case [18], we demonstrate that this summation can be seen as a linear filter. Indeed, for a spatially flat-top beam after a grating, the far-field power P_{FF} is linked to the near-field intensity I_{NF} by the following general expression:

$$P_{FF}(t) = \iint_{\text{Near Field}} I_{NF}[t - \delta t(x, y)] dx dy, \quad (19)$$

where δt is the time delay depending of the location in the beam and the configuration used [19]. Equation (19) is the general expression of the near-field AM spectrum filter. Since the transformation is temporally linear, the AM spectrum is also linearly filtered.

An analytical expression of the transfer function in Eq. (19) can be established in the TSSD configuration for a spatially rectangular flat-top beam. For a plane grating, time delay is given by $\delta t(x) = \frac{x}{h} \Delta T$, with h as the plane grating illuminated length and ΔT as the total time delay. Since Eq. (19) is linear, let us consider a single frequency for intensity modulation f_{AM} and $I_{NF}(t) = 1 + \sin(2\pi f_{AM} t)$ is the near-field intensity before the grating. It leads to the expression of the far-field power

$$P_{FF}(t) = S \left[1 + \text{Sinc}(\pi f_{AM} \Delta T) \sin(2\pi f_{AM} \left(t - \frac{\Delta T}{2} \right)) \right], \quad (20)$$

where S is the illuminated surface. The attenuation factor depends linearly on the AM frequency by the

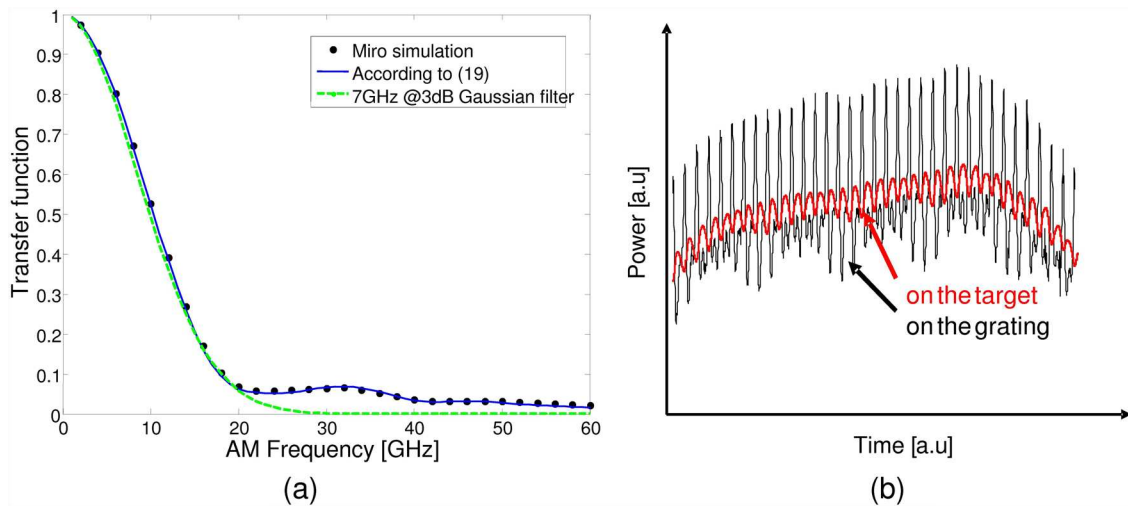


Fig. 13. (Color online) (a) AM spectrum transfer function: comparison between Miró simulations and numerical model according to Eq. (19) and a low-pass Gaussian filter with a 7 GHz bandwidth. (b) Details of temporal shapes before and after the focusing grating in the LMJ configuration (with two sine phase modulations at 2 and 14.25 GHz). It illustrates the AM filtering (no reference level).

function $\text{Sinc}(\pi f_{\text{AM}} \Delta T)$ and the phase modification is given by the sinusoidal term but is only a time delay. It defines analytically the transfer function for this case.

In all the other cases, including the LMJ configuration, we use numerical results given by Miró simulations that we can compare with a MATLAB computation of Eq. (19). This computation is defined by the grating used, the time delay, and the expression of the near field (for a spatially flat-top beam, it only depends on the geometry). Thus, the LMJ configuration transfer function is presented in Fig. 13(a) for both calculations and we have excellent agreement. This AM filter due to focalization on the LMJ configuration reduces the intensity modulation on the target [Fig. 13(b)] and is equivalent to a Gaussian low-pass filter with a 7 GHz 3 dB bandwidth.

We emphasize the fact that this study of the far-field power could be ambiguous; AM is given for the whole focal spot. A local definition of AM should be different. Indeed, because of the speckle pattern that is moving to achieve spatial smoothing, we have for each point of the focal spot a distortion criterion $\alpha = 200\%$, which is obviously inadequate. We made simulations to discriminate both cases. Actually, for a few speckles of spatial integration ($100 \mu\text{m} \times 100 \mu\text{m}$), we are in very close agreement with total far-field values.

The distortion criterion reduction due to this filtering depends on the AM spectral content. As mentioned in Section 2.C, AM spectral content depends on the phase modulation frequencies and the origin of FM-to-AM conversion. Considering first the smoothing modulation only ($f_m = 14.25 \text{ GHz}$), since Eq. (11) is close to a Hermitian function when f_c is small, it almost satisfies Eq. (6) and FM-to-AM conversion due to the crystals' spectral acceptance is mostly at $2f_m = 28.5 \text{ GHz}$. Thus, in the focal plane, the impact of the crystals on the far-field power is

negligible because it is strongly filtered. Simulations for a configuration without grating confirmed that point: before focalization $\alpha_{\text{NF}} = 47\%$ and in the focal plane $\alpha_{\text{FF}} = 3\%$ (Table 2). The second transfer function due to propagation after the plane grating (14) is not Hermitian. So FM-to-AM conversion is mostly at $f_m = 14.25 \text{ GHz}$ and is filtered by the 7 GHz 3 dB bandwidth transfer function. However, it is not negligible: for a frequency conversion configuration with a plane grating, we have $\alpha_{\text{NF}} = 34\%$ and in the focal plane $\alpha_{\text{FF}} = 8\%$. The addition of the anti-Brillouin modulation ($f_m = 2 \text{ GHz}$) changes the AM spectral content and there are harmonics of 2 GHz that are only slightly filtered by the AM filter due to focalization. But as mentioned in Sections 2.D and 3.D, they have no important impact and distortion criterion values are still low. For a configuration without grating, we have $\alpha_{\text{NF}} = 66\%$ and in the focal plane $\alpha_{\text{FF}} = 12\%$. Finally, for the LMJ configuration (two modulations with the plane grating), $\alpha_{\text{NF}} = 39\%$ and in the focal plane $\alpha_{\text{FF}} = 9\%$. These results are summed up in Table 2.

Note that for laser damage of the final optics, α_{NF} should be considered, while α_{FF} should be considered for laser–target interaction as long as at least a few speckles of integration are made.

4. Conclusion

In high-power lasers, FM-to-AM conversion is an important problem for the laser pulse temporal shape control and also for fusion ignition. We have shown that the distortion criterion is not enough to quantify the impact of FM-to-AM conversion and that the AM spectrum has to be analyzed, too. The SCF contribution to FM-to-AM conversion is the most important. We have given analytical models to predict FM-to-AM conversion that are in very good agreement with numerical simulations. With LMJ phase modulations, we have a distortion criterion of $\alpha_{\text{NF}} < 40\%$ before the focusing grating. However, the final focusing

system acts as a linear low-pass-filter on the AM spectral content; AM frequencies higher than 7 GHz are mostly filtered. Finally, we have shown that FM-to-AM conversion in the focal spot is $\alpha_{\text{FF}} < 10\%$, in agreement with the presented models.

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Appendix A

$\tilde{a}_0$ is the phase modulated spectrum given by Eq. (2) and H is a transfer function; we have to consider $\tilde{a}_{\text{out}}$, the filtered spectrum simply defined by $\tilde{a}_{\text{out}}(f) = H(f)\tilde{a}_0(f)$. So

$$\tilde{a}_{\text{out}}(f) = \sum_{n=-\infty}^{+\infty} H(nf_m)J_n(m)\delta(f - nf_m). \quad (\text{A1})$$

The AM spectrum is defined by the intensity Fourier transform

$$\begin{aligned} \tilde{I}_{\text{out}}(f) &= FT[I_{\text{out}}(t)] = FT[|a_{\text{out}}(t)|^2] \\ &= FT[a_{\text{out}}(t)] * FT[a_{\text{out}}^*(t)]. \end{aligned}$$

Using Eq. (A1),

$$\begin{aligned} \tilde{I}_{\text{out}}(f) &= \left[\sum_{k=-\infty}^{+\infty} H(kf_m)J_k(m)\delta(f - kf_m) \right] \\ &\quad * \left[\sum_{n=-\infty}^{+\infty} H^*(nf_m)J_n(m)\delta(f + nf_m) \right], \quad (\text{A2}) \end{aligned}$$

one finds

$$\begin{aligned} \tilde{I}_{\text{out}}(f) &= \sum_{k=-\infty}^{+\infty} \left\{ \sum_{n=-\infty}^{+\infty} H^*(-nf_m)H[(k-n)f_m] \right. \\ &\quad \times J_{-n}(m)J_{k-n}(m) \left. \right\} \delta(f - kf_m). \quad (\text{A3}) \end{aligned}$$

The only frequencies existing in the AM spectrum are harmonics of f_m , so for an integer k , we find Eq. (4):

$$\begin{aligned} \tilde{I}_{\text{out}}(kf_m) &= \sum_{n=-\infty}^{+\infty} H^*(-nf_m)H[(k-n)f_m] \\ &\quad \times J_{-n}(m)J_{k-n}(m). \quad (4) \end{aligned}$$

If k is odd, by rewriting Eq. (4), we have the following expression:

$$\begin{aligned} \tilde{I}_{\text{out}}(kf_m) &= \sum_{n=-\frac{(k-1)}{2}}^{+\infty} \{ [H^*(nf_m)H((n+k)f_m)J_n(m) \\ &\quad \times J_{n+k}(m)] + [H^*(-nf_m)H(-(n+k)f_m) \\ &\quad \times J_{-n}(m)J_{-(n+k)}(m)] \}. \quad (\text{A4}) \end{aligned}$$

Using the property of the Bessel functions $J_{-n}(m) = (-1)^n J_n(m)$ [20], we can rewrite Eq. (A4), which leads to Eq. (5).

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